



VIT-AP
UNIVERSITY

Module 1: Understanding Cloud Computing

Fundamental Concepts and Models

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OUTLINE

- **Fundamental Concepts and Models**
 - Roles and boundaries,
 - cloud characteristics,
 - cloud delivery models,
 - **cloud deployment models.**
- Summary
- References

Cloud deployment models

❑ Cloud deployment models

- A cloud deployment model represents a specific type of cloud environment, primarily distinguished by ownership, size, and access.
- There are four common cloud deployment models:
 - • Public cloud
 - • Community cloud
 - • Private cloud
 - • Hybrid cloud

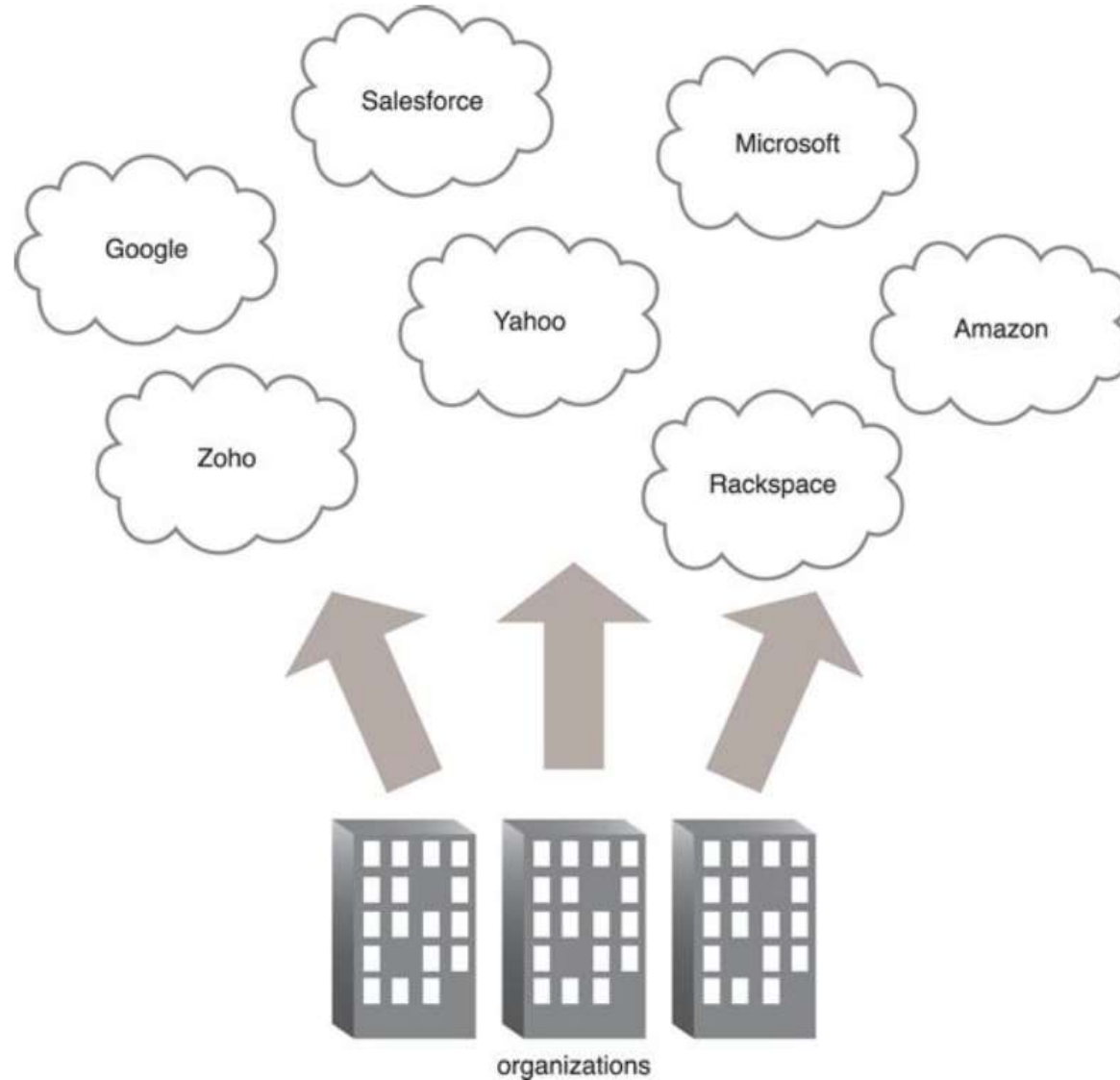
Cloud deployment models

- ❑ Cloud deployment models
- ❑ Public Cloud

- Public Cloud
 - Services offered over the public internet by cloud service providers.
 - These services are available to anyone who wishes to purchase them.
 - Use Cases:
 - Startups and small businesses with limited budgets.
 - Projects requiring scalability and flexibility.
 - Applications with variable workloads.
 - Benefits:
 - Cost-effective: No need to invest in hardware.
 - Scalability: Easily scalable based on demand.
 - Maintenance: Managed by the cloud provider.
 - Challenges:
 - Security: Data is stored on shared infrastructure.
 - Compliance: May not meet specific regulatory requirements.

Cloud deployment models

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- ❑ Public Cloud



Cloud deployment models

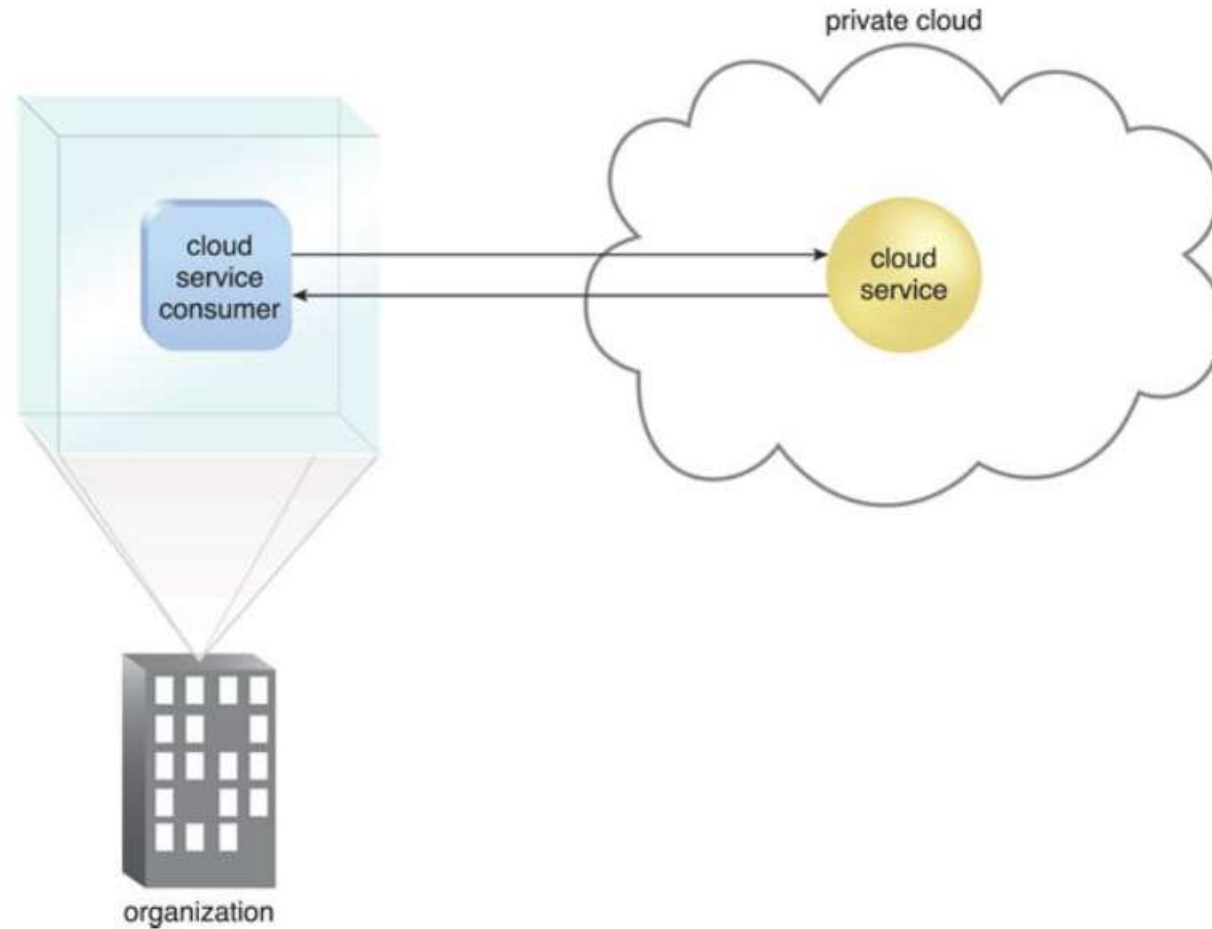
- ❑ Cloud deployment models
- ❑ Public Cloud
- ❑ Private Cloud

- Private Cloud

- Cloud infrastructure operated solely for one organization.
- It can be managed internally or by a third party.
- Use Cases:
 - Large enterprises with specific compliance and security requirements.
 - Organizations needing full control over their data and infrastructure.
 - Industries with strict regulatory requirements, like finance and healthcare.
- Benefits:
 - Security: Enhanced security and privacy as resources are not shared.
 - Customization: Infrastructure can be tailored to specific business needs.
 - Control: Full control over data and resources.
- Challenges:
 - Cost: Higher costs due to investments in hardware and software.
 - Maintenance: Requires in-house expertise for management and maintenance.

Cloud deployment models

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- ❑ Private Cloud



Cloud deployment models

- ❑ Cloud deployment models
- ❑ Public Cloud
- ❑ Private Cloud
- ❑ Hybrid cloud

- Hybrid Cloud

- A combination of public and private clouds, allowing data and applications to be shared between them.

- Use Cases:

- Businesses needing both on-premises resources and public cloud scalability.
 - Applications requiring data privacy but with the need for public cloud resources.
 - Disaster recovery and backup solutions.

- Benefits:

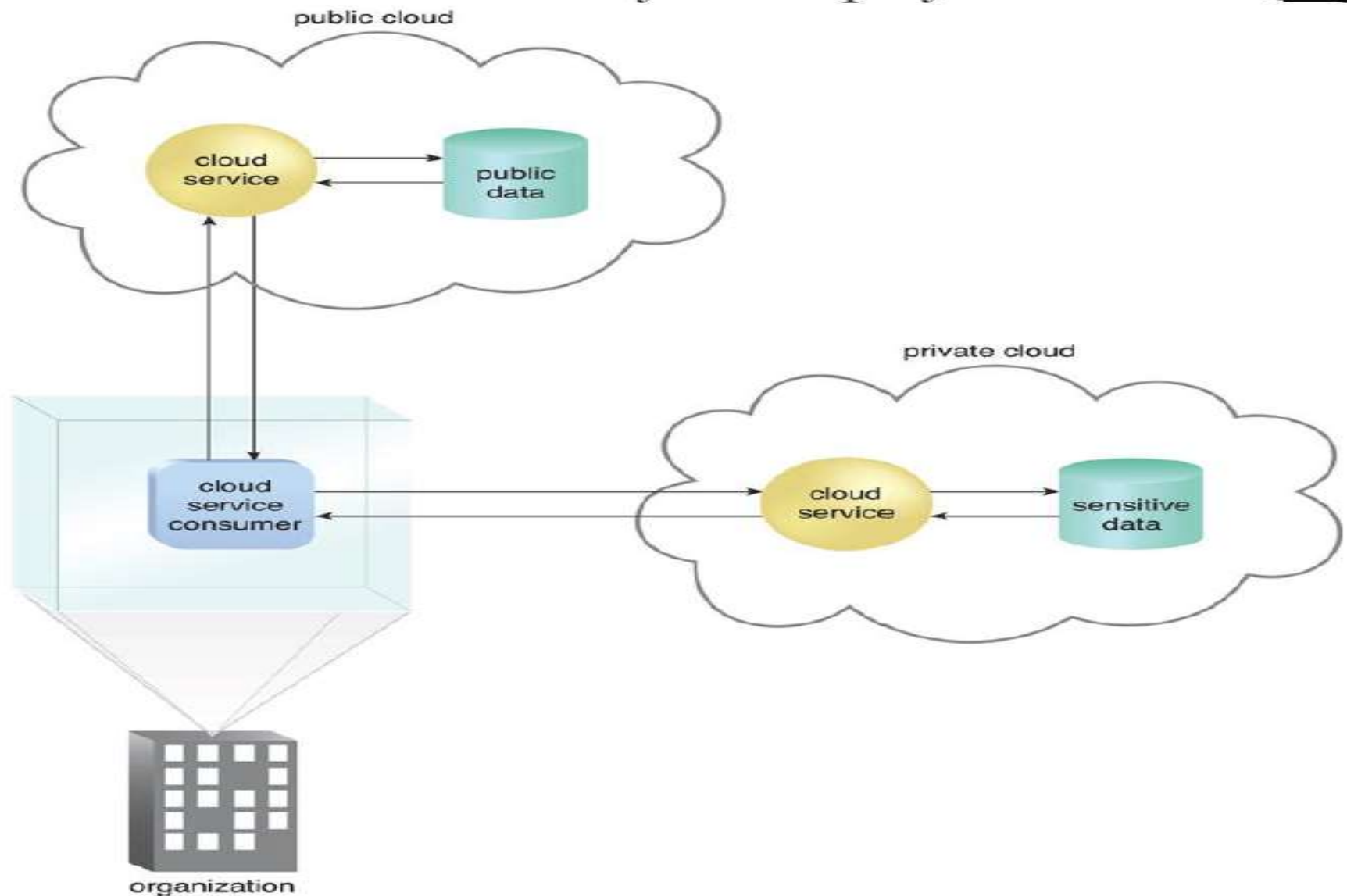
- Flexibility: Allows data and applications to move between private and public clouds.
 - Cost-Efficiency: Balance between cost and security.
 - Scalability: Access to public cloud resources when needed.

- Challenges:

- Complexity: Managing hybrid environments can be complex.
 - Security: Ensuring data security across different environments.

Cloud deployment models

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Cloud deployment models

- ❑ Cloud deployment models
- ❑ Public Cloud
- ❑ Private Cloud
- ❑ Hybrid cloud
- ❑ Community Cloud

- Community Cloud

- A cloud infrastructure shared by several organizations with common concerns, such as security, compliance, or jurisdiction.

- Use Cases:

- Organizations within the same industry or with similar requirements.
 - Collaborative projects or research initiatives.
 - Government agencies or institutions.

- Benefits:

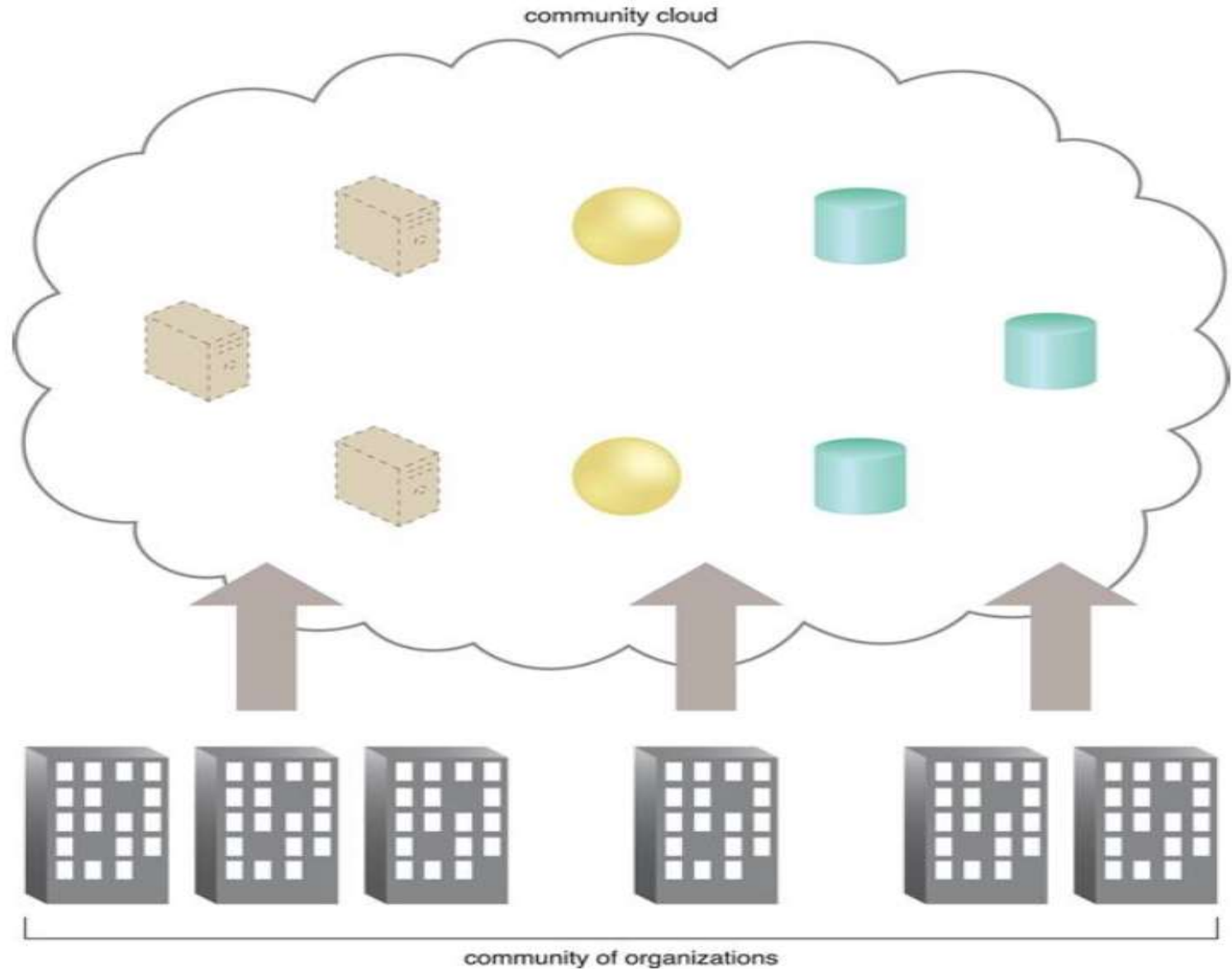
- Cost Sharing: Reduced costs as infrastructure is shared.
 - Collaboration: Facilitates collaboration between organizations.
 - Compliance: Meets specific regulatory or policy requirements.

- Challenges:

- Privacy: Ensuring data privacy between organizations.
 - Management: Coordinating between multiple organizations.

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☐ Cloud Delivery Model

- ☐ Infrastructure-as-a-Service (IaaS)
- ☐ Platform-as-a-Service (PaaS)
- ☐ Software-as-a-Service (SaaS)

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Questions



**Thank
You**